

Shedding Light or Casting Shadows? Relations between Primary and Secondary Cities

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Introduction

Until recently, it was rather common to perceive the hinterland of cities as ‘rural’, directing research to the study of linkages, relations and dependencies between ‘urban’ and ‘rural’ (for example, see Hoggart, 2005). This widespread traditional conception of cities having a predominantly rural hinterland needs to be challenged, as processes of spatial upscaling and expansion have made many cities find other *cities* in their hinterland, suggesting that a focus on regional urban–urban relations is just as valid as a focus on urban–rural relations. Put simply: many functional urban areas or metropolitan areas – terms traditionally used to denote a city and its surrounding commuting area – are nowadays composed of multiple cities. It makes sense to address these regions as multicentric urban areas or as ‘polycentric urban regions’ when these cities do not differ widely in terms of size or overall importance.

Champion (2001) has distinguished three modes through which such multicentric urban regions arise. First, he distinguished a ‘centrifugal mode’: some smaller cities in the sphere of

influence of larger cities may have risen as satellite towns of the latter, as a result of core city expansion and redistribution. Second, an ‘incorporation mode’ is discerned when a larger city extends its sphere of influence to include a formerly rather self-sustaining surrounding town or small city, or a set of several such cities. Third, the ‘fusion mode’ denotes that previously distinct and independent cities of rather similar size become integrated due to the advancement of connective infrastructure and technology, and the spatial extension of activity and travel patterns of people and firms.

Basing our research on European Spatial Planning Observation Network (ESPON) data (projects 1.1.1 and 1.4.3 in particular; see ESPON, 2005, 2011), we can calculate that 1,001 out of the 1,967 European cities are actually located in a multicentric metropolitan area (meaning that they are located in the sphere of influence of another city and vice versa), whereas the others are the sole centre of their functional region. In terms of population, the share is even larger: out of the 265 million inhabitants of these 1,967 cities, almost 192 million people (72.4 per cent) live in a multicentric urban region, while only 73 million people live in cities that do not have other cities in their hinterland. Hence, in sharp contrast to popular conceptions, multicentric urban regions are the dominant mode of urbanization in Europe, not the single city with its more or less suburban commuting area or rural hinterland. This poses an interesting challenge for researchers and policy-makers alike, because many of our urban theories still depart from this outdated notion of city plus hinterland. In policy practice, the complexity of governing multicentric urban regions, with their overlapping boundaries and powers but shared challenges and priorities, needs little emphasis.

With the rise of multicentric urban regions, we also require a framework to capture and describe the unfolding spatial organization of the process of metropolitan integration that they are experiencing. As is widely known in economic theory and in transport geography, increased

integration leads to a rearrangement of economic activities, people and urban functions over space, creating new interactions and interdependencies between the cities making up multicentric urban regions. It is also known that even though there are socio-economic benefits arising from integration (the generative effect), these benefits and the costs of integration do not necessarily spread evenly over a region (a redistributive effect), even when a strong governance framework is put in place (Lefèvre, 1998; Hincks et al, 2017).

Recently, a novel framework for capturing the spatial organization of multicentric urban regions was developed, building on the concepts of ‘borrowed size’ and ‘agglomeration shadows’ (Meijers and Burger, 2017). It is assumed that stronger integration with other cities allows a city to ‘borrow size’: to tap into the agglomeration benefits of nearby cities if they are sufficiently integrated. This could become visible in higher growth rates, or in having more, or more specialized, functions than one would expect given the size of those places. But at the same time, a process of metropolitan integration may lead to cities being in an ‘agglomeration shadow’ when the presence of certain economic activities or urban functions in nearby places means that there is no room to develop these themselves, or when the process of integration empties out these places of higher-order activities and functions since they can be served from other cities. So far, it has remained unclear why one city profits from metropolitan integration in the sense of borrowing size, whereas another may face the agglomeration shadows of nearby cities.

In this chapter we explore and empirically substantiate this framework of borrowed size/agglomeration shadows by positioning cities in a classic multicentric urban region, the Randstad Holland, in the framework developed by Meijers and Burger, and explore which factors are associated with a position in the ‘agglomeration shadow’ or the ‘borrowing size’ quadrants. It is not our intention to establish causality at this early stage of exploration, which would also allow

robust policy recommendations, but rather to seek general characteristics that typify primary and secondary cities, and hence, what differentiates them, and the challenges they face. These different points of departure are important to avoid ‘one-size-fits-all’ urban development strategies, not just with respect to the differences between primary and secondary cities, but also taking the heterogeneity among secondary cities into account. In the process, we will also test the borrowed size/agglomeration shadow framework and ultimately develop it further.

The theory section goes deeper into the process of increased integration between cities and its associated benefits and costs – so-called ‘network externalities’. Subsequently, we expand the current ‘borrowed size/agglomeration shadow’ framework and present an empirical exercise that leads to the classification of places in the Randstad into this framework. This classification is then associated with six factors, whose relevance we explain later in this chapter: historical importance, size, sectoral composition of the local economy, social status, network embeddedness and tourism. Finally, the conclusion synthesizes our findings and sketches a research agenda that emerges from these results.

Theory

The basic idea that further integration between cities yields important benefits is widespread. It would lead to what Capello (2000) has termed ‘urban network externalities’. This concept refers to the benefits originating from relationships between places (see also Camagni, 1993; Meijers, 2005; Boix and Trullén, 2007; Van Oort et al, 2010; Burger and Meijers, 2016) and can be described as external economies from which firms and households can benefit by being located in agglomerations that are well embedded or positioned in networks that connect with other agglomerations.

While an emerging scholarship on the benefits of such networks between cities is developing, the simple importance of relationships between cities has been stressed many times in historical accounts of the evolution of the European urban system (for example, De Vries, 1984; Hohenberg and Lees, 1995). The idea that ‘the spatial organisation in which cities operate is fundamental to understanding their efficiency, growth, factor productivity and sometimes their specialisation’ (Capello and Camagni, 2000: 1484) is also at the heart of the urban systems literature that had its heyday in the 1960s, 1970s and early 1980s (Pred, 1977; Bourne and Simmons, 1978).

Currently, we are witnessing a revival of such scholarship as there are five different schools studying relationships between cities (Peris et al, 2018): one predominantly focusing on the regional or intra-metropolitan scale and centred on concepts of polycentricity (for example, Kloosterman and Musterd, 2001; Parr, 2004); one addressing the global scale with a focus on world city firm and communication networks (see Alderson and Beckfield, 2004; Taylor and Derudder, 2016); one somewhat diffuse school rooted in urban economics (‘the new economic geography’) employing economic modelling approaches (for example, Fujita et al, 1999; Partridge et al, 2008); one school employing simulation and complexity theories to understand the behaviour of individual and institutional agents building the urban system from the bottom up, often drawing inspiration from physics (see Batty et al, 1999; Bretagnolle and Pumain, 2010); and, finally, one seeking regularities (‘laws’) with respect to city size distributions within an urban system (see Gabaix and Ioannides, 2004; Bettencourt et al, 2007). Despite their many ontological and epistemological differences, these approaches share a focus on functional and economic types of relationships and networks between cities.

Despite this attention, it is hard to quantify or precisely define the importance of network embeddedness for individual cities. While studies repeatedly show the importance of a city's embeddedness in large-scale networks of all kinds – firms, capital, knowledge, people, goods – for its performance (Taylor, 2003; Bel and Fageda, 2008; Neal, 2013; Meijers et al, 2016), how such networks' importance compares to local factors is largely unknown. Some time ago, scholars argued that local factors still seem to substantially outweigh network effects (Boix and Trullen, 2007), but it appears that the importance of network externalities is growing as the connectivity of places has been rapidly increasing in the past decades. McCann and Acs (2011) argue that nowadays national and international connectivity is more important for performance than size. This was empirically substantiated by Meijers et al (2016), who found that the presence of particular metropolitan functions was better explained by connectivity than by local size – for example, international organizations, headquarters, advanced producer services, banks, fairs, conferences, patents. It would seem that networks substitute for proximity (Johansson and Quigley, 2004), but this could also be a more complementary relationship – particular local conditions may, for instance, also foster a good network embeddedness. The nature of the interaction between local factors and network factors is unknown, and in this light, this chapter associates local factors with network relatedness and ultimately with the outcome of network integration.

Some studies have shown the presence of such network externalities in the particular context of multicentric urban regions (see Maly, 2016). It was found, for instance, that the more closely cities were integrated, the more network externalities were present, while the lack of integration between nearby cities would even lead to negative urban network externalities (Meijers and Burger, 2017). Recently, it was shown that higher levels of functional integration and institutional integration (metropolitan cooperation) contribute to the presence of more

agglomeration benefits at the scale of a network of cities within a single urban region (Meijers et al, 2018).

However, while positive urban network externalities arise at the level of urban regions as a whole (or, perhaps better put, at the level of the network of cities), it remains a question whether individual cities profit from metropolitan integration by definition. It could well be that a generative effect at the level of urban regions hides an intra-regional distributive effect. Such effects have been established in transportation studies and in economic theory, most notably the New Economic Geography. In transportation studies this has been referred to as ‘the two-way road argument’ (Preston, 2001: 18), meaning that new infrastructure (that is, stronger integration) between a core and peripheral region can promote outward investment just as easily as inward investment. The core–periphery models of the New Economic Geography suggest that integration expands market areas, but also leads to more competition between firms in the periphery and in the core city. The latter are often producing more advanced products, while the former are losing their local monopoly. The outcome of integration depends on the balance between centripetal and centrifugal forces (Krugman, 1991; Fujita et al, 1999), and transportation costs are a central element in determining this balance. Low costs make it possible to serve the region from one place, so it then makes sense to profit from agglomeration economies in the largest place. The presence of such distributive effects of new infrastructure has been shown by, for instance, Meijers et al (2012).

Local politicians and policy-makers often pay lip service to the potential generative effects of city network integration, but are probably most interested in the distributive effects. The simple fact that the number of functions and activities in any multicentric urban region is not infinite, and that their distribution basically is a zero-sum game, means that cities compete heavily with other

cities in their multicentric region, certainly if their tax revenues are mostly comprised of local taxes.

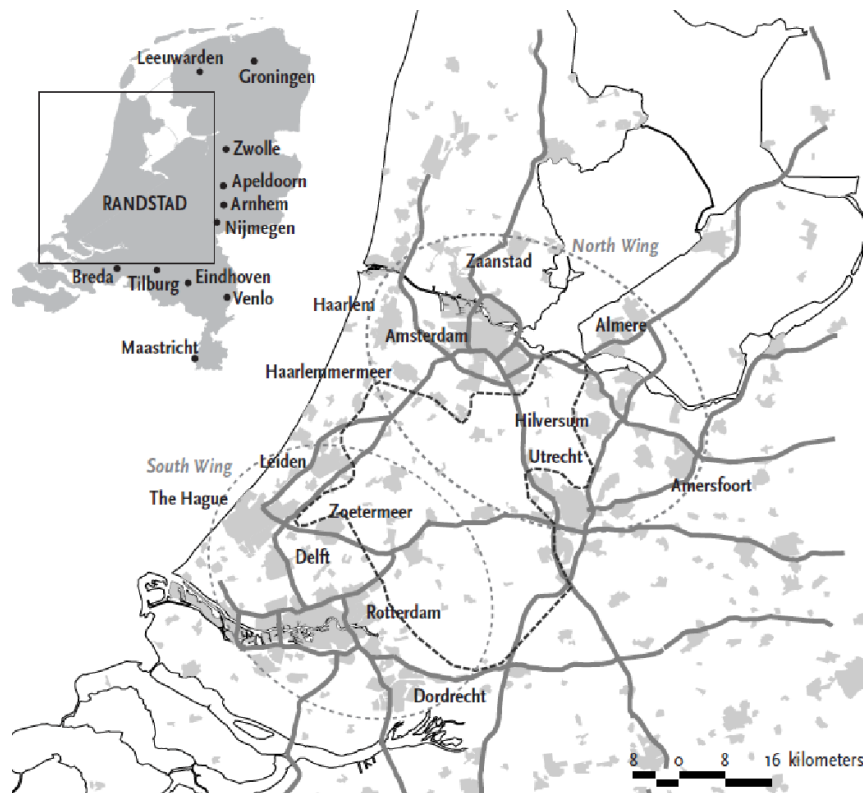
Recently, a conceptual framework was introduced that captures the possible outcomes of further integration in multicentric urban regions (Meijers and Burger, 2017). It builds on two concepts: ‘borrowed size’ (Alonso, 1973) and ‘agglomeration shadows’ (appropriated from the New Economic Geography literature, for example Krugman, 1993). Size borrowing occurs when a city exhibits urban functions and/or performance levels normally associated with larger cities that enjoy corresponding agglomeration externalities, and this pattern is a consequence of being integrated with those cities. Agglomeration shadows are essentially the opposite effect, yet are also the product of integration with nearby or more distant cities. It could be said that a place is under an agglomeration shadow when it has fewer functions and lower performance levels than expected given its size, because of competition effects from cities with which it is integrated through networks and relations. Volgmann and Rusche (2020) draw an interesting parallel with the regional concept of spread and backwash effects, conceptualized by Myrdal (1957).

Classifying cities in the Randstad

In this chapter we explore why one city profits from metropolitan integration in the sense of borrowing size, whereas another may face the agglomeration shadows of nearby cities. We present a case study of the Randstad Holland region, a classic example of a multicentric metropolitan region (see Hall, 1966; Kloosterman and Musterd, 2001). We will first classify cities by positioning them in the ‘borrowed size/agglomeration shadow’ framework. Then, we will seek for factors that can be associated with this classification.

The Randstad is not just a multicentric region, as the rather equal size distribution of its main cities (Amsterdam, Rotterdam, The Hague, Utrecht) also makes it polycentric. It may seem that this means that there are no secondary cities in the region, but the opposite is true. There are many other medium-sized cities (for example, Haarlem, Leiden, Dordrecht, Delft, to name but a few) as well as numerous smaller ones. There is no precise or formal delimitation of the Randstad – meaning the population estimate varies between 7 and 8 million inhabitants – and only about a third of this population lives in the four main cities; thus, the presence of the secondary cities is very important and the region is truly multicentric. Some proxy the area by the four provinces it covers, but we adopt a more commonly agreed upon delimitation that excludes the (former) island of Goeree-Overflakkee, the Noordoostpolder, most of the north of Holland (for example Alkmaar) and the Noordoostpolder in Flevoland province. ‘Our’ Randstad had 7.46 million inhabitants in January 2018 (see Figure 2.1). Generally, the cities in the Randstad all have distinct histories and identities and they share a long trajectory of being related to and also competing with each other. Only a few places can be considered suburban new towns – but such ‘growth poles’, stemming from the 1980s, have also developed their own identities that fuel, compete with and are related to the larger neighbouring city that once acted as their metropolis in the proper Greek meaning of that word (the mother city from which settlers came). Our unit of analysis will be the 120 municipalities in the Randstad.

Figure 2.1: The Randstad Holland



These municipalities will be positioned in the ‘borrowed size/agglomeration shadow’ quadrant matrix, which requires us to assess, first, the connection between their size and the urban functions present in the city. Second, we explore the connection between their size and their performance in terms of urban dynamics. We proxy urban functions with two variables: the number of jobs (2016; data from Statistics Netherlands) and the presence of restaurants (2011, data from the Bedrijfschap Horeca en Catering). We used 2011 data, as this measurement allowed us to compare the development of restaurants over the 1996–2011 period. Several studies have used restaurants as a proxy for the presence of agglomeration benefits, their number to proxy advantages of scale and their variety to explain advantages of scope (Schiff, 2015; De Vos and Meijers, 2019). Here, performance was proxied with the population growth (2002–18) and the development of the number of restaurants (1996–2011).

Obviously, one of the omissions of the current ‘borrowed size/agglomeration shadow’ framework is that many places do not have more or less urban functions than expected given their size, or a development trajectory that is better or worse: many places simply have the expected number of functions or development dynamics. As such, the four quadrant framework needs to evolve into a nine square matrix that allows for this classification. Each of the nine squares can be labelled according to its most defining characteristic. The two squares in the upper right can both be titled ‘borrowed functions’ and the two in the bottom left both classify as ‘borrowed performance’. Figure 2.2 presents this extended framework.

Figure 2.2: The extended ‘borrowed size/agglomeration shadow’ framework

Dimensions of borrowed size		Connection size $\leftrightarrow$ function	
		Fewer functions than expected given size	More functions than expected given size
Connection size $\leftrightarrow$ performance	Performs less well than expected given size	<i>Agglomeration shadow</i>	<i>Borrowed functions</i>
	Performs better than expected given size	<i>Borrowed performance</i>	<i>Borrowed size</i>



Dimensions of borrowed size		Connection size $\leftrightarrow$ function		
		Fewer functions than expected given size	As many functions as expected	More functions than expected given size
Connection size $\leftrightarrow$ performance	Performs less well than expected given size	<i>Agglomeration shadow</i>	<i>Performance shadow</i>	<i>Borrowed functions</i>
	Performs as expected	<i>Functional shadow</i>	<i>As expected</i>	<i>Borrowed functions</i>
	Performs better than expected given size	<i>Borrowed performance</i>	<i>Borrowed performance</i>	<i>Borrowed size</i>

To position the cities, we calculated the number of restaurants and jobs per capita for the 120 municipalities in the Randstad. The average number of restaurants per 1,000 people was 0.55 (standard deviation (*SD*) = 0.30), with a minimum of 0.17 and a maximum of 2.47. The average number of jobs per 1,000 people was 40.11 (*SD* = 15.55), with values ranging from 17 to 107. Those places that had jobs and restaurants above the mean + 0.75 *SD* were considered to have more functions than expected. Likewise, those that had greater than 0.75 *SD* below the average value, were considered to have fewer functions than expected.

A similar procedure was applied to explore whether population growth and development of functions (restaurants) was above or below the expected value. The average population growth in the Randstad region was 8.4 per cent over the 2002–18 period (*SD* = 10.79), and the average increase in restaurants was 9.69 per cent (*SD* = 29.32). Again, we used deviations in excess of 0.75 *SD* to determine which places performed better or worse than expected.

One complication with this procedure is that in quite a few cases a place performs better according to one indicator, but worse according to another that measures the same dimension. For instance, a place may have more jobs than expected, but less restaurants than predicted for its size. In those cases, if a place scores better or worse on one indicator, and as expected on the other, we categorized it as scoring better or worse. In the unusual case that a place scored better on one indicator and worse than expected according to the other, we classified it as ‘as expected’ on this dimension, unless the average standardized score was still above 0.75 *SD* (categorized as ‘better than expected’) or below 0.75 *SD* (worse than expected). This resulted in the following classification of Randstad municipalities (Figure 2.3).

Figure 2.3: Classification of municipalities in the Randstad

Dimensions of borrowed size		Connection size ↔ function		
		Fewer functions than expected given size	As many functions as expected	More functions than expected given size
Connection size ↔ performance	Performs less well than expected given size	Agglomeration shadow Dronten, Maassluis, Nissewaard, Strijen, Oegstgeest, Kaag en Braassem, Papendrecht, Hardinxveld-G., Oostzaan, Binnenmaas, Korendijk, Oudewater, Haarlemmerliede en S.	Performance shadow Blaricum, Huizen, Dordrecht, Montfoort, Veenendaal, Zederik, Bloemendaal, Vlaardingen, Soest	Borrowed functions Zoeterwoude, Ouder-Amstel, Laren, Gorinchem
	Performs as expected	Functional shadow Rhenen, Landsmeer, Heemskerk, Zuidplas, Alblasterdam, Voorschoten, IJsselstein, Hellevoetsluis, Heemstede, Lopik, Eemnes, Molenwaard, Wormerland, Leiderdorp, Krimpen a/d IJssel	As expected Ridderkerk, Cromstrijen, Zoetermeer, Amstelveen, Leerdam, Lelystad, Zaanstad, Nieuwkoop, Zwijndrecht, Capelle a/d IJssel, Uithoorn, Oud-Beijerland, Baarn, Hillegom, Woerden, De Bilt, Beemster, Westvoorne, Brielle, Giessenlanden, Noordwijkerhout, Bodegraven-R., Wassenaar, Weesp, Krimpenerwaard, Purmerend, Katwijk, Alphen a/d Rijn, Teylingen, Schiedam, Gooise Meren, Stichtse Vecht, De Ronde Venen, Velsen, Waterland, Gouda	Borrowed functions Amsterdam, The Hague, Westland, Beverwijk, Renswoude, Zeist, Haarlem, Hilversum, Wijdemeren, Delft, Lisse, Noordwijk, Utrechtse Heuvelrug, Slidrecht, Zandvoort, Nieuwegein
	Performs better than expected given size	Borrowed performance Pijnacker-N., Houten, Lansingerland, Uitgeest, Albrandswaard, Hendrik-Ido-Ambacht, Almere, Wijk bij Duurstede, Castricum, Leidschendam-Voorburg	Borrowed performance Midden-Delfland, Bunschoten, Bunnik, Waddinxveen, Edam-Volendam, Leusden, Woudenberg	Borrowed size Barendrecht, Diemen, Aalsmeer, Utrecht, Vianen, Haarlemmermeer, Rijswijk, Amersfoort, Leiden, Rotterdam

Explaining patterns in the Randstad

Research approach

In what follows we will explore which factors correlate with whether a place is able to borrow performance, functions, size or, instead, faces a performance or functional shadow. In this explorative phase, we will consider six possible factors using descriptive methods to explore their association with our classification.

First, it can be assumed that path dependency plays a role. Places that were traditionally important centres for their region may have retained that historically established position, even though other places may have grown faster. For each contemporary municipality, we identified the largest city and considered its population size in 1849. The first census in the Netherlands is from this year. We subsequently constructed an indicator that gives the rank of a city in 1849. The higher ranked (so, the lower the number), the more historically important a city is.

Second, we simply assume that size plays a role. In line with previous findings (Burger et al, 2015; Meijers et al, 2016), the general pattern seems to be that larger cities more often borrow size, while smaller places are more likely to experience an agglomeration shadow. We use population size for 2016.

Third, we consider the sectoral composition of the local economy, which has long been known (Echevarria, 1997) to lead to unbalanced growth (Gardiner et al, 2013). Here, we calculated the ratio between white-collar workers (employees in the public and private services sector) and blue-collar workers (employees in construction and manufacturing). We assume that high scores are associated with function-rich places. Data on sectoral composition of the economy of municipalities is available through Statistics Netherlands, and refers to the year 2016.

Fourth, we include the social status score. This is a composite measure reflecting residents' level of education, position in the labour market and income, and is calculated by the Netherlands Institute for Social Research. Since it is calculated at the neighbourhood spatial scale, we calculated a population-weighted average at the municipal level. Data is for 2016. We assume that places with a lower social status are more often in the shadow of other places.

Our fifth factor appeals quite literally to the idea that cities can borrow size in networks (Meijers et al, 2016), and that networks may substitute for the benefits of agglomeration (Johansson and Quigley, 2004). We created a network embeddedness factor to find out whether places that are more strongly related to other places do indeed perform better or have more functions than expected. In other scientific fields, the co-occurrence of phenomena in texts is used as an indicator of their relatedness, and likewise we can assume that when two cities are often mentioned in the same breath, they are related to each other, more so than cities that are hardly mentioned together. Important to understand is that this conceptualization of relatedness is broader than connections or physical flows between cities. Our measure of relatedness is based on the co-occurrence of Dutch place names in all (25 million) websites with the .nl extension (the most commonly used extension in the Netherlands) – see Meijers and Peris (2019) for its exact calculation. The absolute pattern of relationships, based on the frequency of co-occurrences, was explained by applying the gravity model. This allowed us to gain a relative understanding of the importance of a relationship. For instance, Rotterdam and Amsterdam are the cities that are most often mentioned together on Dutch websites, but you also expect this to be the case given the sheer size of both cities and the limited distance between them. If we correct for the size of the places, and the distance between them, by using the gravity model, we are able to compare predicted strengths of relationships between places with actual strengths (co-occurrences). Predicted values

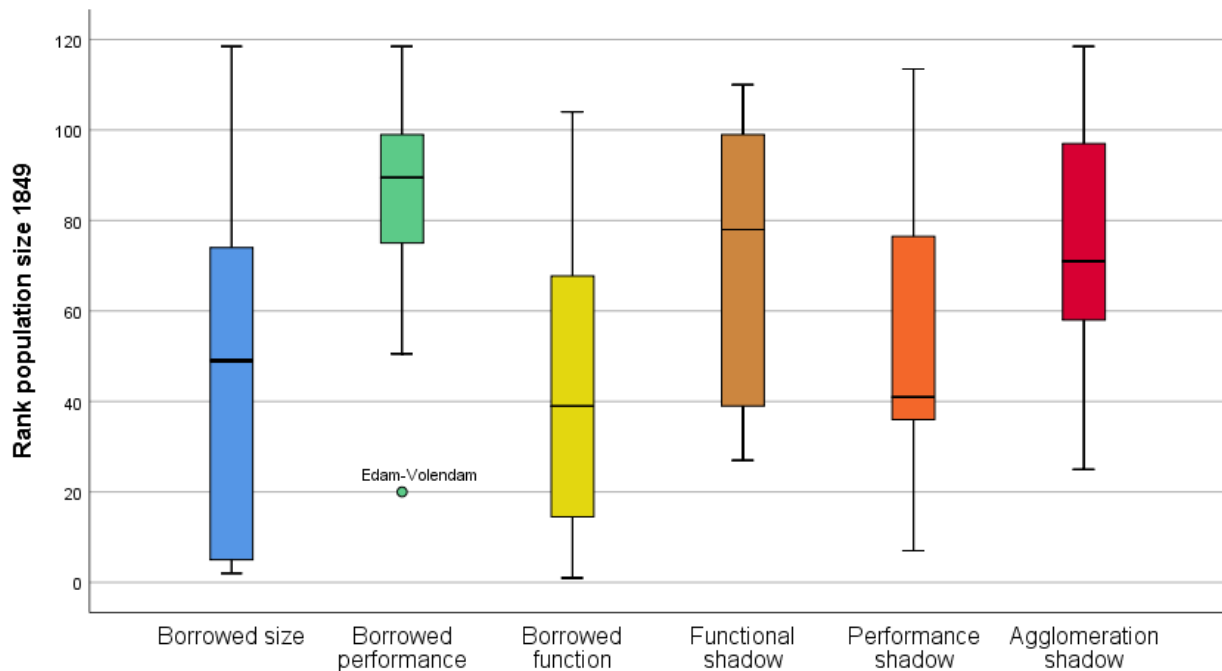
of less than 0, which happened for pairs of small villages far from each other, were set at 0. Our indicator gives the aggregated difference between all predicted values of the strength of the relationship with almost 1,700 other Dutch places, and the actual value. To move from place level to municipal level, we approximated the municipal network embeddedness with the relationships of the place that gave its name to a municipality, and when no such place name was used, we took the largest place in that municipality. In the handful of cases where the municipal name is composed of two place names, we took the average values.

Sixth, a clear example of a means through which size can be borrowed is tourism, as tourists temporarily add to the size and hence support base for functions, while this sector has also grown considerably in many places. Our ‘tourism’ indicator gives the number of hotel beds per capita in 2011 – data is derived from the Bedrijfschap Horeca en Catering.

Results: aggregate patterns

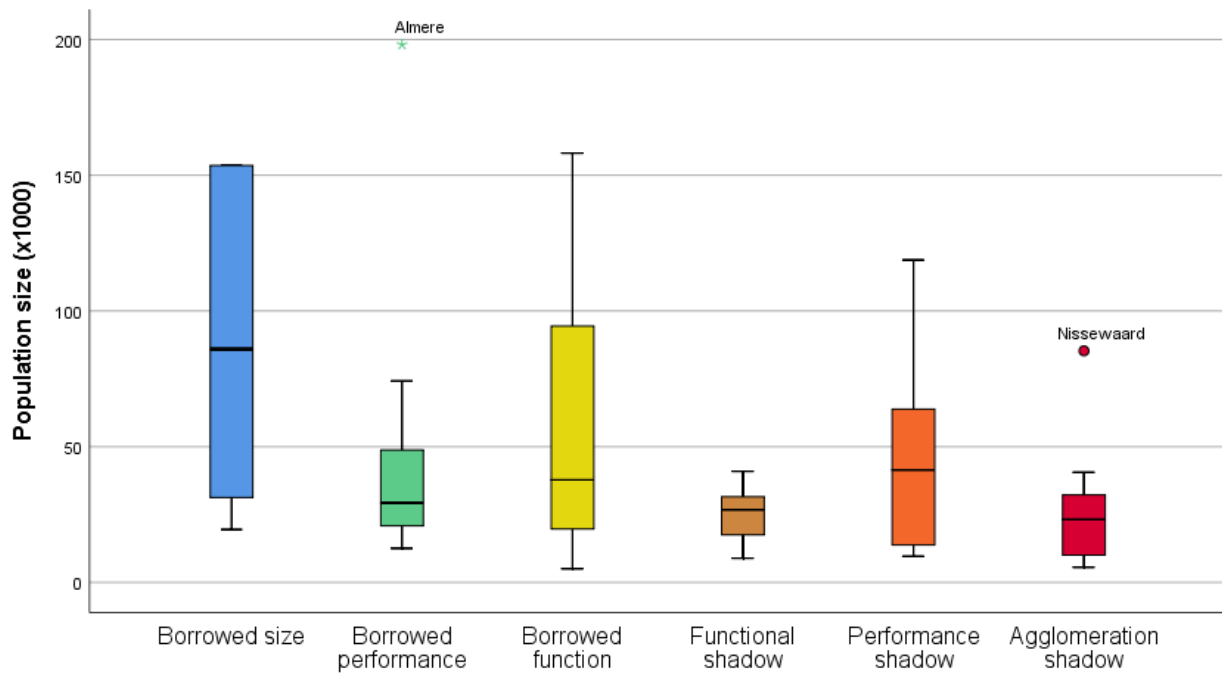
We explore the association between six categories of our ‘borrowed size/agglomeration shadow’ framework and these six factors through a series of box plots (Figures 2.4–2.9; for visibility, we leave out the ‘as expected’ category). The colours used refer to the colours given to these categories in Figures 2.2 and 2.3.

Figure 2.4: Historical importance



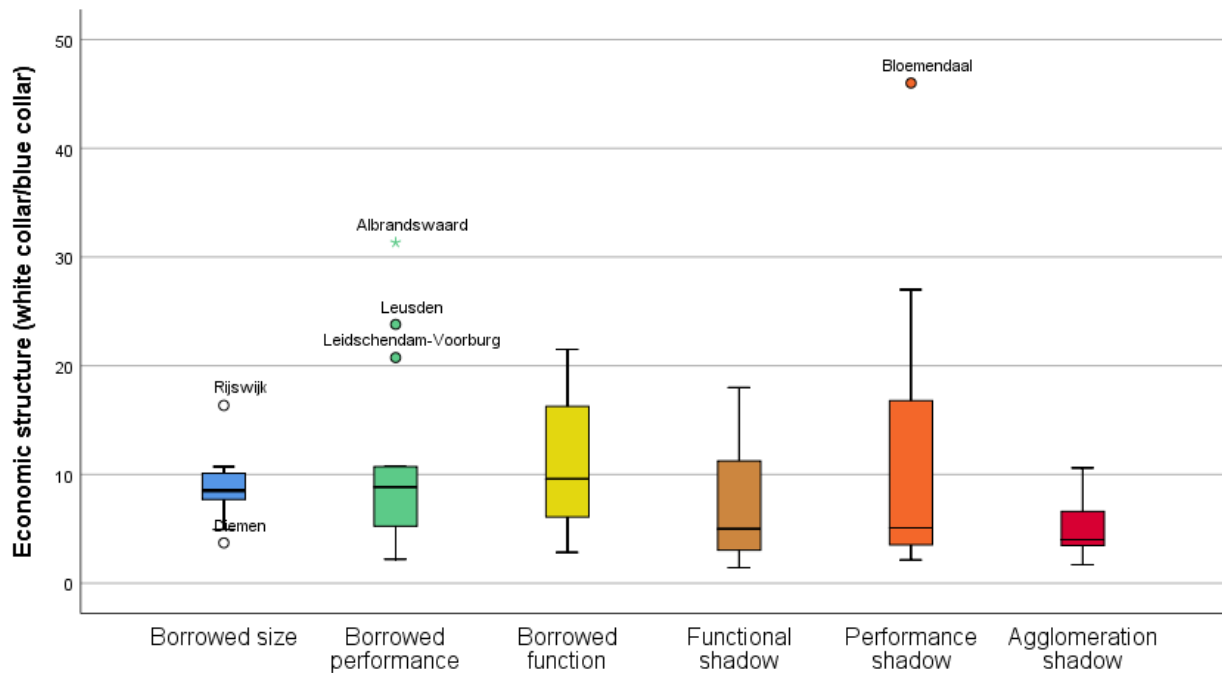
The pattern regarding the historical importance of places is quite clear. Places that were important 170 years ago tend to be relatively more represented in the ‘borrowed size’ and ‘borrowed function’ category. Most of them seem to have kept their functional centrality and, consequently, we don’t find the most historically important places in a ‘functional shadow’ or in the ‘borrowed performance’ category. The historically most important place that is now in an agglomeration shadow is Maassluis – this former fishing town close to Rotterdam seems to have become overshadowed by its larger neighbour. On the other side, Haarlemmermeer was totally insignificant in 1849 (ranked 119 out of 120), and now borrows size. It is the municipality close to Amsterdam where Schiphol Airport is located – an obvious case of borrowing size indeed. Places that borrow performance tend to be historically less important. Hence, their growth may be explained more in terms of suburbanization – actually several growth centres are included here: for example, Almere, Houten, Lansingerland, Pijnacker-Nootdorp (see Figure 2.5).

Figure 2.5: Population size



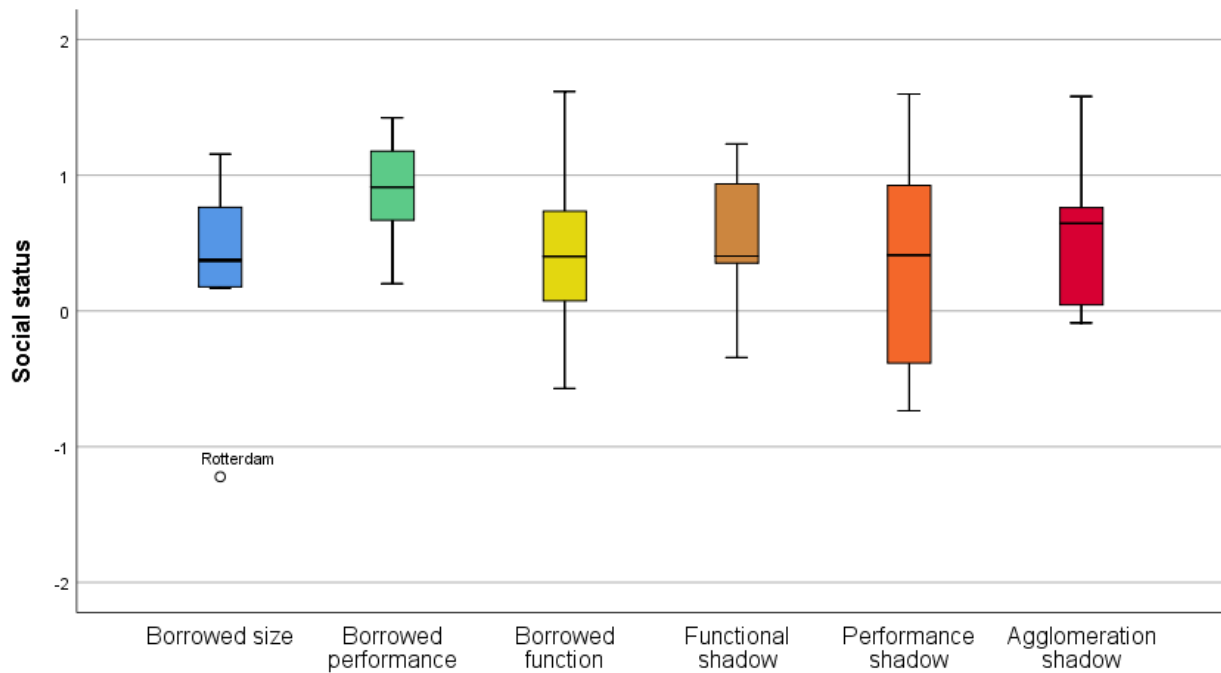
Turning to contemporary importance in terms of size, it is quite obvious that larger cities are more often in the borrowed size or the borrowed functions category. Almere is growing rapidly and can be considered in many ways a suburb of Amsterdam, but its position in the borrowed performance class means that it has not been able to develop functional importance (yet). Smaller cities are more often in a functional or agglomeration shadow. Nissewaard, including the city of Spijkenisse, is also not able to live up to the expectations created by its size – this is an example of one of the earlier growth poles near Rotterdam that grew rapidly due to industrialization in the 1970s and 1980s but has stagnated in more recent years – perhaps in many ways similar to the ‘first suburban ring’ in the US.

Figure 2.6: Economic structure



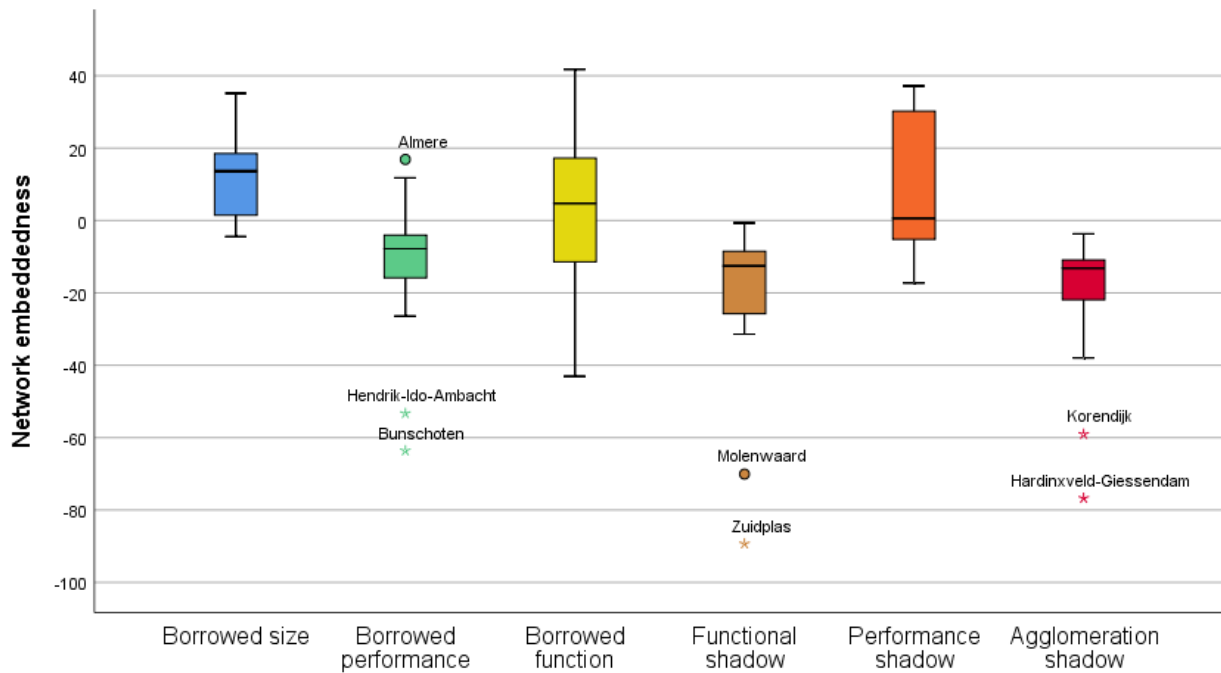
The economic structure does play a role in positioning cities in our framework (see Figure 2.6). Those in which manufacturing is stronger, and which have relatively fewer services, tend to experience an agglomeration shadow more often, while the median values of places in performance and functional shadows suggest the same for those categories. The local economy of places that ‘borrow’ tends to be more characterized by services.

Figure 2.7: Social status



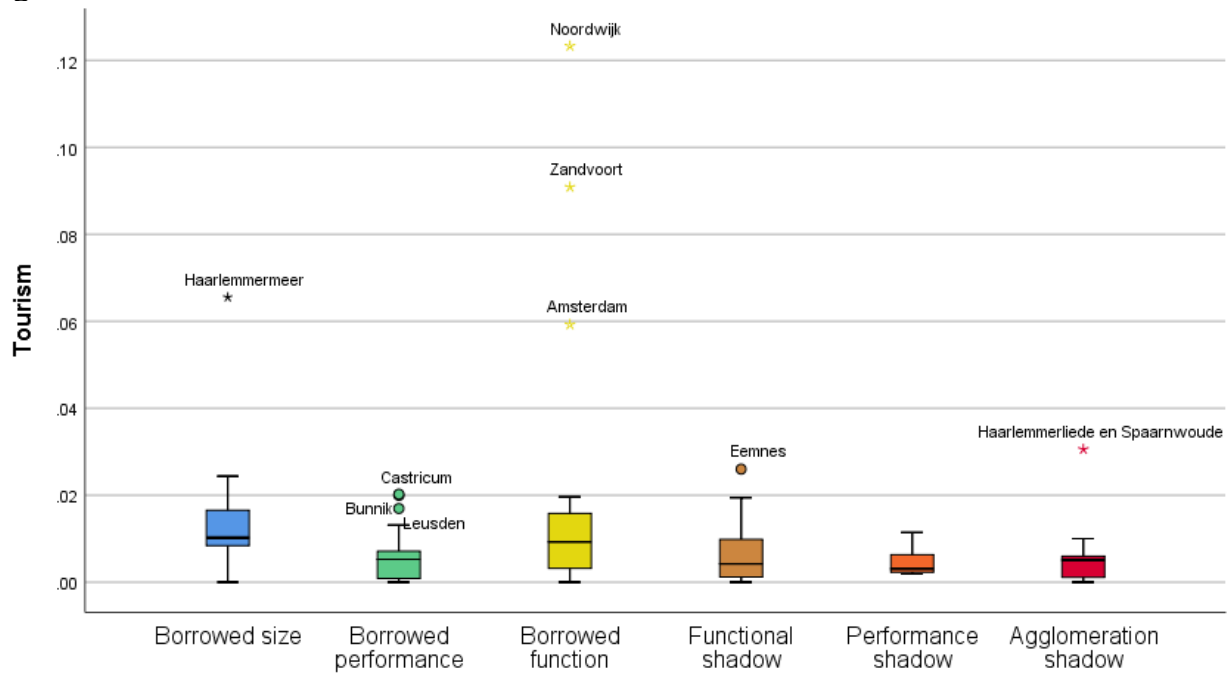
In terms of social status, the pattern is rather mixed – or simply absent (see Figure 2.7). The social status of places in the agglomeration shadow does not seem to differ too much from those that borrow size. In fact, the place with the lowest social status – Rotterdam – is borrowing size. Noteworthy perhaps is that places that are growing more tend to have a somewhat higher social status, while quite a few municipalities that are in a performance shadow are among the socially weaker. A link can be established with the fact that those places in the borrowed performance category include more recent suburban areas to which relatively affluent families tend to migrate.

Figure 2.8: Network embeddedness



The boxplot for network embeddedness is interesting from various perspectives (see Figure 2.8). First of all, the places that borrow size are much more related to other places, whereas at the opposite end, those that are in an agglomeration shadow have far fewer relationships with other places. Second, places that borrow function also seem to be able to do so because they have stronger relationships. Third, places in a performance shadow also tend to be slightly more related to other places. This seems to call for further analysis in the types and directions of relationships, an issue that needs to be taken on board when the toponym co-occurrences method is further developed. Their opposite – the places that borrow performance – are actually less strongly related to other places. We had expected that places in a functional shadow would also have stronger relationships with other places, to make up for the lack of functions locally, but this is not the case.

Figure 2.9: Tourism



Tourism adds, temporarily, to the size of a place, and is one of the clearest examples of borrowed size from elsewhere (see Figure 2.9). It obviously has a positive impact on the functions in a place, as the borrowed size and borrowed function categories are more touristic. However, it does not lead to borrowed performance. Places in a shadow – whatever type – tend to be less touristic. Note that our indicator – hotel beds – does not just refer to touristic places; places that are central in business networks also pop up. The clearest example is Haarlemmermeer (Schiphol).

Results: functional dimension

Figure 2.10 presents the association between our factors and whether places have more or fewer functions than expected given their size. When comparing municipalities that have more functions (N = 30) with those that have fewer functions (N = 38), it turns out that municipalities that punch above their demographic weight are those that are: (a) historically important: $t(66) = -4.0, p = .000$;

(b) larger: $t(30) = 2.5, p = .017$; (c) have a lower social status: $t(66) = -2.2, p = .033$; and (d) draw more tourists: $t(32) = 2.4, p = .022$. Somewhat counterintuitive is that places with a lower social status manage to have more functions than expected. Perhaps this is because function-rich larger cities are often also home to lower social classes, in addition to a comparatively smaller upper class. While places that do better also seem to have a more services-oriented economy, this difference between ‘more’ and ‘fewer’ functions was not statistically significant. Finally, and perhaps most strikingly, places with stronger relationships with other places tend to have more functions: $t(66) = 4.9, p = .000$. This provides an important empirical underpinning for the proposition that networks substitute for size.

We also ran an ordinal regression model explaining the score on the functional dimension, and including all six covariates simultaneously. In this model, tourism and network embeddedness turned out to be significant factors ($p < 0.05$), with historical importance being significant only at the $p < 0.10$ level.

Results: performance dimension

When zooming in on just the performance dimension, associations with most of the six factors become clearer (Figure 2.11). In comparison to places that underperform, places that grow faster seem to be: (a) historically more important; (b) larger; (c) have a higher social status; (d) have more service sector jobs; (e) have more visiting tourists; and (f) are more strongly related to other places. However, comparing the mean scores for the ‘better’ and ‘worse’ categories renders only the difference in size significant, $t(28) = 2.1, p = .046$. The difference in tourism is significant only at the $p = .094$ level, $t(50) = 1.7$. Note that there are just 27 municipalities in the ‘better’ category and 26 in the ‘worse’ category, while at least $N = 30$ is recommendable for independent samples

t-tests. All in all, this performance dimension shows weaker differences between the categories in comparison to the functional dimension.

An ordinal regression model explaining the score on performance and including again the six covariates, identified social status as the most significant factor ($p = 0.028$), followed by population size ($p = 0.052$) and network embeddedness ($p = 0.091$).

Figure 2.10: Borrowed functions

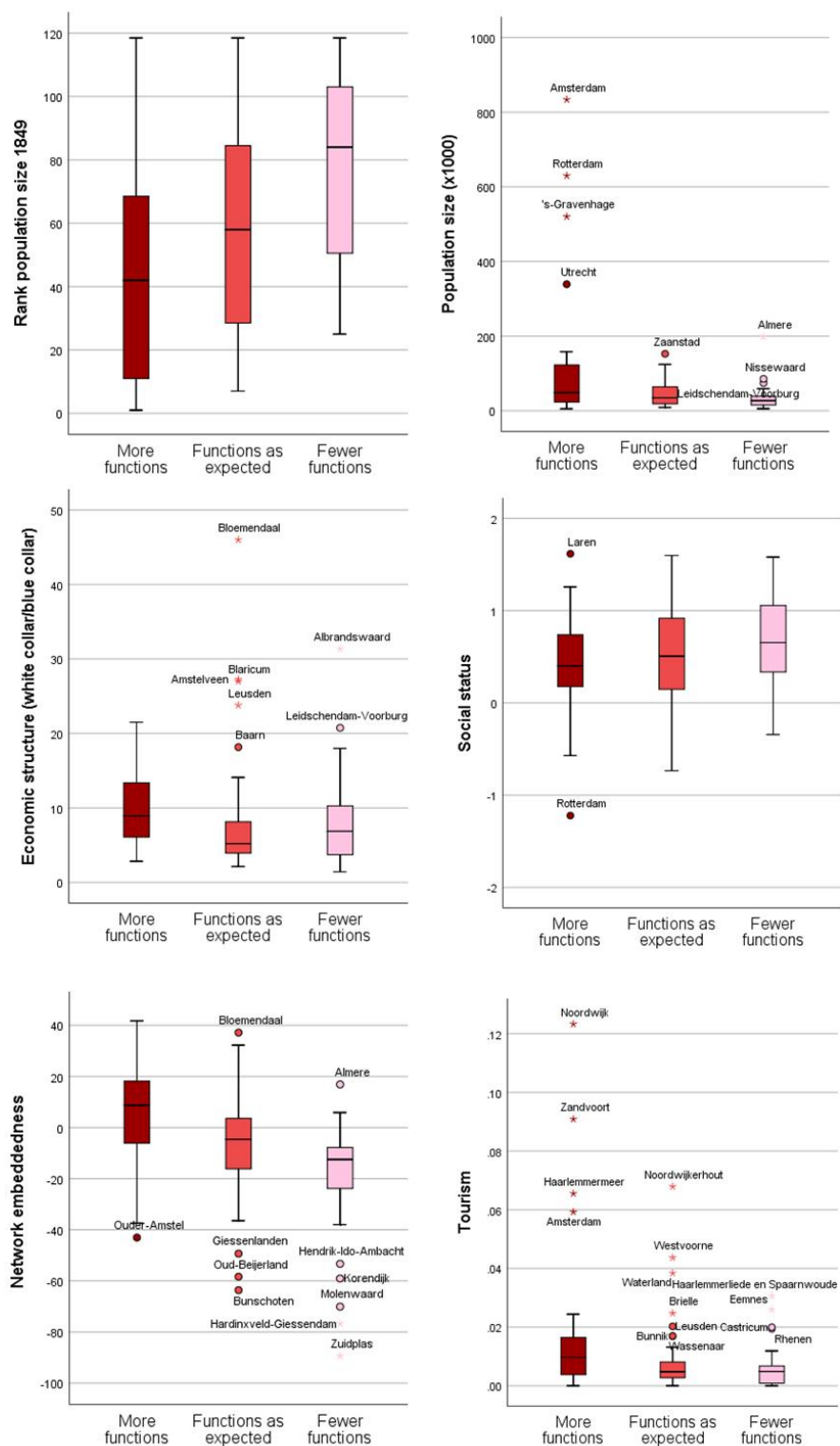
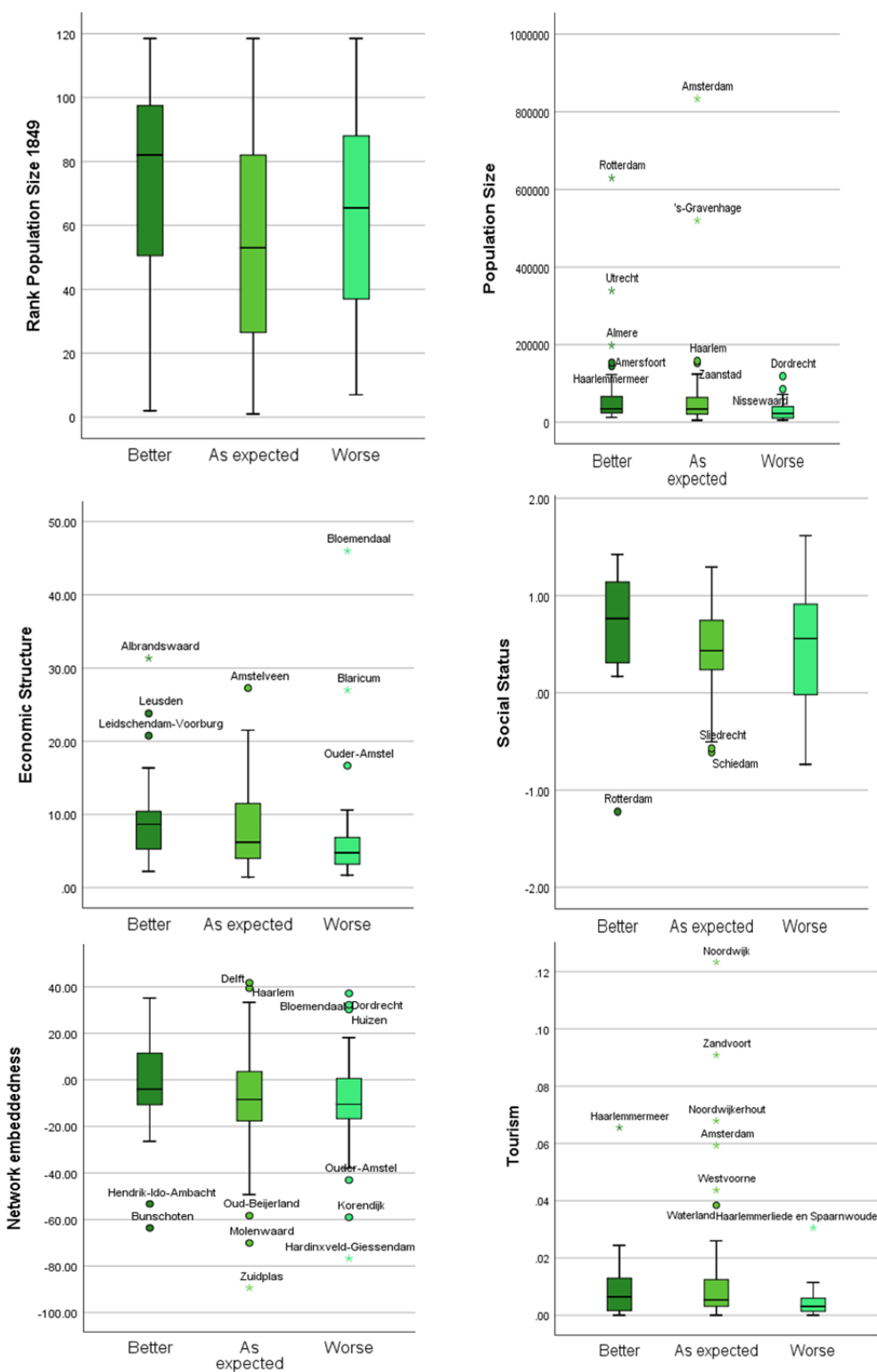


Figure 2.11: Borrowed performance



Conclusion

While much attention has been devoted to understanding city network externalities, as a substitute and/or complement to agglomeration externalities, research has focused much less on how the costs and benefits of the further integration of cities in networks are distributed over the cities in the network of cities. The ‘borrowed size/agglomeration shadow’ framework (Meijers and Burger, 2017) proposed previously to identify the different outcomes of network integration at the scale of individual cities was taken as a point of departure in this chapter. We have proposed a further refinement of this original four quadrant matrix, and now identify seven possible local outcomes of network integration. Cities can: (1) experience an agglomeration shadow; (2) experience a performance shadow; (3) experience a functional shadow; (4) perform as expected; (5) borrow functions; (6) borrow performance; and (7) borrow size.

This elaboration of possible local outcomes was further explored and tested for the Randstad Holland region, a classic multicentric and polycentric urban region in the Netherlands. Each of the 120 municipalities in this metropolitan region (7.5 million inhabitants) was positioned in the elaborated ‘borrowed size/agglomeration shadow’ framework. In terms of the secondary cities that resemble the definition used in this book (see Chapter 1) most are positioned in very different quadrants. Some face a ‘performance shadow’ (Dordrecht, Vlaardingen), some perform as expected (Schiedam, Zaanstad), while others ‘borrow functions’ (Delft, Haarlem, Hilversum) and two manage to ‘borrow size’ (Amersfoort, Leiden). While it is noteworthy that none of these cities has fewer functions than expected, these varied positions at the same time suggest that the group of ‘secondary cities’ is quite heterogeneous, as is also discussed in Chapter 5, this volume. The ‘first cities’ either borrow functions (Amsterdam, The Hague) or borrow size (Utrecht, Rotterdam).

Given this heterogeneity, it makes sense to explore how local conditions are associated with the positioning of cities. In this chapter, we associated six factors with this positioning: historical importance, size, sectoral composition of the local economy, social status, network embeddedness and tourism. All factors seem to play a role in the positioning. This role is sometimes easier to understand when looking at the functional and performance dimensions of the ‘borrowed size/agglomeration shadow’ framework separately.

Places that have more functions than expected are associated with being historically important, and also with being more populous, while this population on average has a lower social status. But most particularly, they draw relatively more tourists (a very clear example of borrowing size of course) and they are much more strongly related to other cities than places that have fewer functions than expected. This provides clear support for the idea that networks substitute for size.

Places that witnessed stronger development since the late 1990s, are, in comparison to places that underperformed, characterized by a greater population size, while they also have a higher social status and draw more tourists, and more relations with other cities (network embeddedness). The boxplots in Figure 2.11 presented suggest some evidence for the two remaining factors.

While processes of further integration are positive for the region as a whole, the benefits and costs are not spread evenly over primary and different types of secondary cities. Our results suggest that local circumstances and path dependencies may influence the capacity to exploit processes of integration in multicentric metropolitan areas. One could argue that cities have different roles to play in such a region, after all, the essence of networks is that they allow for specialization. But this does require insight into the new interdependencies and a regional planning

strategic framework to make the most of the possibilities of metropolitan network integration, especially if we are to ensure that costs and benefits of this process are spread more evenly.

It may have been noted that we refer to those relationships as associations and not as causal relationships. We believe that this fits better with the explorative nature of the empirical analysis here presented. But the issue of causality is of great importance, and in many ways deserves further careful analysis. This analysis may be more a matter of discussing the evolution of secondary cities in relation to first cities in the region than of mere quantitative statistical analysis to establish causality, since historical pathways are likely to play a key role. The city of Dordrecht, for instance, was definitely one of the most important Dutch cities in the 16th and 17th centuries. But it has gradually been overshadowed by Rotterdam, which, among other factors, enjoyed a more favourable location for trade and political prioritization, translating into more public investment in the last century when its large harbour and associated industrial complexes were considered the economic engine of the Netherlands. Of course, the growth and decline of the most important economic sectors are likely to play key roles, as is perhaps sheer coincidence. For instance, Schiphol airport, in the last decades another engine of economic growth, would probably have been relocated from its current position near Amsterdam to a place near Leiden and The Hague by the end of the 1930s if the war with Germany had not loomed on the horizon. Therefore, through such historical accounts, we may get an understanding of why some cities possess the factors that seem to allow them to borrow size, whereas others lack them.

Importantly, such accounts will also reveal that there are shifts in the positioning of some cities – those that borrow size nowadays may have faced agglomeration shadows in the past. It is very possible that we will witness a shift in the coming decade (2030) – the quick rise of agglomeration costs in the largest cities (rents, congestion, ‘touristification’ of city centres) already

leads to many spillover effects in nearby places, a process likely to be further aggravated by the need for space during pandemics like COVID-19. There is one important difference, though, with previous patterns in suburbanization. The households and firms that are attached to the urban way of life do not move to nearby ‘bedroom communities’ that lack urban charm, but ‘hop over’ to secondary cities that tend to be well connected and rich in functions and that have historical urban character and identity – an emerging process in the Netherlands that we could perhaps call ‘hop-over urbanization’. Consequently, in addition to our ‘snapshot’ that was based on a rather limited number of indicators to capture the functional and performance dimensions of each city, further research could adopt a dynamic perspective and try to explain shifts in positions of cities, thereby building on a more comprehensive analysis of: (a) proxies for functions and performance; and (b) a larger set of factors that can be associated with these two dimensions. It is not unlikely that this leads to the observation that cities have simultaneous and overlapping experiences with borrowing and shadowing, demanding a more careful account of their position.

The issue of causality is also pertinent when it comes to the role of relationships between pairs of cities and the networks these shape at the scale of the urban system. Do these relationships develop because of the many functions a city has, or do these relationships indeed substitute for size and allow a city to sustain more functions? Both explanations are obviously possible, and, for instance, an instrumented variables approach may add some clarity here. Also, there are many opportunities to further elaborate the network embeddedness variable, such as considering network embeddedness on different spatial scales or, in the case of secondary cities, the levels of relationship with just the first city. There are many exciting research avenues possible that can build on, and further develop, the approach and the framework presented and explored here.

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